

Carbohydrate Polymers 신규 게재 논문 트렌드 분석 리포트

리포트 생성 일시: 2026-07-22 21:57 UTC | 분석 대상 신규 연구논문: 89편

1. 개요

본 리포트는 Elsevier 저널 Carbohydrate Polymers(ISSN 0144-8617)에 새로 게재된 연구논문 중, 이전 회차에서 이미 다룬 논문을 제외한 신규 연구논문 89편을 대상으로 제목과 초록을 분석하여 작성되었습니다. 리뷰 논문(review/systematic review 등) 및 미생물 자체의 생물학적 특성·균주 개발·배양 최적화가 핵심 주제인 논문은 수집 단계에서 제외하였습니다.

2. 트렌드 분석

2-1. 소재별 분포 (뜨고 있는 주제/키워드 경향)

수집된 신규 연구논문 89편의 제목·초록을 키워드 매칭으로 분석한 결과, 셀룰로오스 계열과 전분 계열이 각각 19편, 19편으로 가장 활발히 연구되는 소재로 나타났고, 키토산(16편)이 뒤를 이었습니다. 펙틴과 덱스트란/덱스트린 계열도 각 9편으로 꾸준히 등장하며, 히알루론산 기반 생체의료용 소재 연구(6편)도 두드러집니다.

구분	논문 수	비중
셀룰로오스/유도체 (Cellulose)	19	21%
전분 (Starch/Amylose/Amylopectin)	19	21%
키토산 (Chitosan)	16	18%
펙틴 (Pectin/Pectic)	9	10%
덱스트란/덱스트린 (Dextran/Dextrin)	9	10%
알지네이트 (Alginate)	8	9%
히알루론산 (Hyaluronic acid)	6	7%
해조류 다당류 (Fucoidan 등)	4	4%
프락탄/이눌린 (Fructan/Inulin)	3	3%
겔란검 (Gellan gum)	3	3%
카라기난 (Carrageenan)	2	2%
세균 유래 EPS (Exopolysaccharide)	2	2%

2-2. 응용 분야 경향

응용 측면에서는 나노입자/나노복합체 소재화와 생분해성·친환경 소재 개발이 가장 큰 비중을 차지했으며, 항균·항박테리아 기능을 부가한 상처 드레싱/코팅재, 자가치유(self-healing) 및 주사형(injectable) 하이드로겔을 이용한 약물전달·상처치유 응용이 뚜렷한 트렌드로 확인됩니다. 장내미생물 조절 효과를 매개로 한 다당류의 생리활성 연구도 꾸준히 보고되고 있습니다.

구분	논문 수	비중
나노입자/나노복합체	21	24%
생분해성/친환경	17	19%
항균/항박테리아 응용	16	18%
에어로겔/폼	10	11%
약물전달/서방출	9	10%
캡슐화/봉입 (Encapsulation)	8	9%
필름/코팅재	8	9%
장내미생물 조절	6	7%
자가치유 소재 (Self-healing)	6	7%
상처치유 (Wound healing)	6	7%
항암/항종양	4	4%
주사형 하이드로겔	3	3%

2-3. 공통적으로 수행된 필수 분석 기법

구조 및 물성 규명을 위해 다수 논문에서 반복적으로 사용된 분석 기법을 집계한 결과는 아래와 같습니다. 특히 FTIR과 분자량 분석이 다당류의 화학적 골격 및 사슬 특성을 확인하는 기본 절차로 가장 널리 쓰였고, 겔·필름형 소재에서는 유변학(rheology) 측정과 팽윤율(swelling) 측정이 물성 평가의 핵심 지표로 활용되었습니다. NMR은 다당류의 정밀한 결합 구조(단당류 조성, 글리코시드 결합 위치) 규명에 널리 쓰였습니다.

구분	논문 수	비중
분자량 분석 (Molecular weight)	12	13%
팽윤율 (Swelling)	9	10%
FTIR (적외선분광)	8	9%
NMR	7	8%
유변학 측정 (Rheology)	6	7%
SEM (주사전자현미경)	5	6%
XRD (X선회절)	5	6%
DSC (시차주사열량)	3	3%
제타전위 (Zeta potential)	2	2%
TGA (열중량분석)	2	2%

2-4. 논문 구조/포맷의 공통 패턴

- ① 서론에서 대상 다당류/소재의 유래(식물·해조류·미생물 발효 부산물·농산 부산물 등)와 산업적/의학적 필요성을 제시
- ② 추출·정제 또는 화학적 개질(가교, 치환, 복합화) 방법을 상세히 기술
- ③ FTIR·NMR·XRD 등 구조 분석으로 화학 구조/결정성을 확인
- ④ SEM/TEM으로 미세구조(morphology)를 관찰하고, TGA/DSC로 열적 안정성을 평가
- ⑤ 유변학적 물성, 팽윤율, 기계적 강도 등 기능적 물성을 측정
- ⑥ 응용 목적에 맞춘 기능성 평가(항균력, 약물 방출 거동, 세포/동물실험, 흡착·에너지 성능 등)로 마무리

2-5. 전체 공통점

① 대부분의 논문이 천연 유래 다당류의 지속가능성(sustainability)을 핵심 가치로 내세우며 석유화학 소재 대체를 지향합니다. ② 다기능화(multi-functionalization) 전략 — 단일 소재에 항균성, 자가치유성, 전도성, 반응성(자극 감응형) 등 복수의 기능을 동시에 부여하는 설계가 압도적으로 많습니다. ③ 구조-물성-기능의 3단 논리 전개(structure-property-function)가 거의 모든 논문에서 공통적으로 나타납니다. ④ 생체의료(상처치유, 약물전달, 조직재생)와 환경/에너지(흡착, 에너지저장, 방열, 담수화) 두 응용 축으로 연구가 크게 양분되는 경향을 보입니다.

3. 집중 분석: paper / cellulose / pectin / hydrogel / okara / paper coating / biomass / soybean protein isolate 관련 논문

위 키워드에 해당하는 논문은 총 39편으로, 이번 회차 신규 논문의 44%를 차지합니다. 금번 수집 결과에는 okara, paper coating, soybean protein isolate(SPI) 관련 논문은 확인되지 않았으며, cellulose·hydrogel·pectin·biomass·paper(기질) 관련 논문이 다수를 차지했습니다.

3-1. 셀룰로오스(cellulose) 관련 논문 (16편)

Cationic architecture engineering of amphiphilic ionic liquids for synergistic cellulose processing: From molecular interaction to high-performance regenerated materials.

Developing efficient solvent systems to overcome cellulose recalcitrance is pivotal for its valorization. While ionic liquids show promise, the role of cationic architecture in modulating solvation mechanisms remains underexplored. This study investigates a series of amphiphilic ionic liquids (ALLs) with evolving cationic cores, from monocyclic imidazoles to bicyclic superbases (DBN, DBU), to elucidate their impact...

Uniaxial tensile strain-insensitive and zero-birefringence cellulose acetate films via atom-economic isocyanate modification.

Zero-birefringence protective films are essential for polarizers in flexible displays, preventing light leakage and color shifts. However, conventional films, such as cellulose triacetate, often suffer from residual birefringence induced by solvent evaporation and mechanical stress. Here, we employ a facile, atom-economical carbamate reaction between cellulose acetate and p-tolyl isocyanate (PTSI) to produce a...

Lightweight and eco-friendly methyl cellulose foams featuring moisture-induced form interlocking: A versatile alternative to conventional plastic foams.

The plastic pollution crisis demands sustainable alternatives to petroleum-based foams. Cellulose-based foams, renewable and biodegradable, are a good candidate. In this study, fully degradable cellulose foams are fabricated from methyl cellulose (MC) alone via a simple, additive-free method. The method combines mechanical foaming and oven drying without blowing, stabilizing, or cross-linking agents. First, stable...

Skin-inspired cellulose-based hydrogel design with hydrophobic-conductive synergy for enhanced electrochemical nanoplastic removal.

In aquatic environments, traditional cellulose-based hydrogels face structure change challenges due to excessive swelling caused by hydrophilic groups, resulting in poor conductivity and limiting their electrochemical applications. To overcome these challenges, this study presents a bioinspired carboxymethyl cellulose/polyacrylamide/butyl acrylate/polyaniline (CMC/PAM/BA/PANI) composite hydrogel, mimicking the smart...

Antifreeze moisture-retaining coating based on ionic dextrin-hydroxyethylcellulose hydrogels: An initial insight into the structural and functional properties.

The development of antifreeze and moisture-retaining materials is important for cold-chain packaging, outdoor infrastructure, and agricultural protection. In this study, three Dextrin-hydroxyethyl cellulose (HEC) hydrogels with distinct crosslinking strategies were investigated: H1, a borate-acrylic acid covalently crosslinked network; H2, a poly(3,4-ethylenedioxythiophene)polystyrene sulfonate (PEDOT:PSS)-modified...

Geometry-matched chelation enhances Li⁺ diffusion on electrodes binded by modified carboxymethyl cellulose.

Carboxymethyl cellulose (CMC) is attracting increasing interest as a sustainable binder for flexible lithium-ion batteries, yet its dense H-bond network and poorly organized oxygen functionalities limit both chain mobility and efficient Li⁺ coordination. Here, we report a molecular-engineering strategy in which CMC is grafted with flexible, low-polarity polyisoprene chains and subsequently oxidized to generate...

Hydroxypropyl cellulose/thermoplastic polyurethane blend films with reversible optical performance triggered by water.

As the demand for sustainable materials increases, cellulose-based additives have attracted significant attention for use in polymer blends, particularly in applications requiring environmentally responsive behavior. In this work, hydroxypropyl cellulose (HPC)/thermoplastic polyurethane (TPU) blend films were fabricated by a one-step casting method and systematically investigated for their water-triggered optical...

Co-solvency phenomenon and thermodynamic mechanism of cyanoethyl cellulose in acetone-water binary solvent.

Cyanoethyl cellulose (CEC), which is insoluble in both water and acetone individually, exhibits unexpected solubility in acetone-water mixtures—a phenomenon that has not been previously reported or mechanistically explained. Herein, we investigate this anomalous co-solvency behavior by analyzing solution transmittance alongside Huggins and Kraemer parameters to evaluate solvent quality. The results reveal that with...

Synergistic dual-dynamic-bonds rendered adaptable oxidized carboxymethyl cellulose/lignin/sodium alginate/poly(vinyl alcohol) biohydrogels for wound healing.

Dual dynamic bond hydrogel dressings, known for their exceptional self-healing abilities and stimulus responsiveness, offer significant benefits in promoting wound healing. Hence, a multifunctional hydrogel (OCMC/LA/AP/PVAE) was developed by forming dynamic imine and borate ester bonds among oxidized carboxymethyl cellulose, aminated lignin, modified sodium alginate, poly (vinyl alcohol) and...

Robust, hydrophobic and antimicrobial TiO₂-decorated bamboo cellulose nanofiber aerogel with honeycomb porous structure for highly efficient radiative cooling.

Passive radiative cooling is a potentially sustainable thermal management strategy amid escalating global climate change. However, non-renewable radiative cooling materials often suffer from insufficient environmental durability and poor mechanical properties, which limit their practical applications. Therefore, there is an urgent need to develop renewable thermal management materials for radiative cooling that...

Sustainable and reusable antibacterial aerogels constructed from surface-quaternized cellulose nanofibrils for effective oil-water separation.

Severe environmental risks from oil extraction leaks and solvent-containing wastewater have created an urgent need for efficient oil-water separation materials. Cellulose nanofibril (CNF) aerogels are regarded as promising candidates for oil-water separation owing to their low weight, high porosity, and excellent adsorption capabilities. However, the limited surface functionality of CNF restricts their development...

Production of electrospun cellulose fibres via water-assisted rapid deacetylation: Experimental and molecular modelling insights.

Cellulose acetate (CA) enables the production of cellulose nanofibers via electrospinning, overcoming the solubility limits of native cellulose. However, rapid regeneration of CA into cellulose remains a bottleneck for scalable manufacturing. We evaluated three regeneration methods: immersion, vapour exposure, and spray, targeting near-instant deacetylation of electrospun CA fibres. Immersion was the most effective:...

Ultralight core-shell polypyrrole@bacterial cellulose aerogel for broadband electromagnetic wave absorption and efficient heat insulation.

The rapid expansion of high-power electronics and wearable systems exposes them to complex electromagnetic and thermal stresses. External electromagnetic interference (EMI) and heat in the environment cause marked surface warming, a compound effect that accelerates material degradation and performance decline, thereby compromising long-term device reliability and safety. To address this dual challenge, this study...

Pros and cons of wood cellulose binders for 4.6 V LiCoO₂ lithium-ion batteries.

Raising the charging cutoff voltage of LiCoO₂ cathode is an effective route to boost the energy density of lithium-ion batteries. However, when operated above 4.5 V, LiCoO₂ electrode suffers from accelerated structural degradation and severe interfacial side reactions, leading to rapid capacity fading. Conventional PVDF binders offer limited protection under such high-voltage conditions due to their weak interfacial...

Photothermal polypyrrole-cellulose nanofiber membranes as advanced platforms for osmotic energy conversion.

Using nanofluidic membranes to harvest osmotic energy at the interface between seawater and river water is a promising solution for mitigating energy shortages. However, the development of stimuli-responsive ion transport into nanofluidic membranes remains largely underexplored. Herein, we design a photothermally responsive nanofluidic membrane by integrating polypyrrole (PPY), TEMPO-oxidized cellulose nanofibers...

Reinforced clusteroluminescent cellulose nanospheres by strengthening hydrogen bonds networks.

Understanding the strengthening process of the hydrogen-bond network in cellulose is of great significance for enhancing clusteroluminescence performance of the non-conjugated clusteroluminescent cellulose nanomaterials. In this study, clusteroluminescent cellulose nanospheres were synthesized from pulp fibers through a controlled enzymatic-acid hydrolysis process, allowing for the precise analysis of particle...

3-2. 하이드로겔(hydrogel) 관련 논문 (19편)

Laponite-reinforced marine bioinspired injectable self-healing adhesive chitosan-based hydrogel for infected wounds.

Standard dressings struggle to conform to irregular wounds, which can easily lead to bacterial infection and impair the healing process. In this study, an injectable, self-healing hydrogel inspired by mussel and oyster adhesion was developed from dihydrocaffeic acid-modified chitosan, oxidized sodium alginate, and Laponite nanosheets. Benefiting from the synergistic reinforcement of dynamic imine bonds and the...

Skin-inspired cellulose-based hydrogel design with hydrophobic-conductive synergy for enhanced electrochemical nanoplastic removal.

In aquatic environments, traditional cellulose-based hydrogels face structure change challenges due to excessive swelling caused by hydrophilic groups, resulting in poor conductivity and limiting their electrochemical applications. To overcome these challenges, this study presents a bioinspired carboxymethyl cellulose/polyacrylamide/butyl acrylate/polyaniline (CMC/PAM/BA/PANI) composite hydrogel, mimicking the smart...

Sodium alginate modified by dopamine multifunctional injectable hydrogel with photothermal, antibacterial, and hemostatic properties for accelerating infected wound repair.

Sodium alginate (SA)-based wound dressings have attracted increasing attention for wound management. In this study, a multifunctional injectable alginate hydrogel was developed via a dopamine modification strategy. Dopamine-grafted sodium alginate (SA-DA) was synthesized and combined with polydopamine nanoparticles (PDA) to obtain an SA-DA/PDA hydrogel with strong tissue adhesion and photothermal functionality. In...

On-demand amorphous calcium phosphate-mediated regeneration of dentin and enamel using ion-regulated hydroxypropyl methylcellulose and carboxymethyl chitosan hydrogels.

This study presents an innovative heteropolysaccharide hydrogel system engineered through the synergistic combination of hydroxypropyl methylcellulose calcium (HPMC-Ca) and carboxymethyl chitosan-phosphate (CMCS-P). This dual-network hydrogel uniquely leverages the complementary physicochemical properties of both polysaccharides to achieve simultaneous biomimetic mineralization of enamel and dentin-addressing a...

Chitosan-gelatin hybrid hydrogel incorporating strontium-doped ceria nanoparticles for enhanced wound healing and mitochondrial function recovery.

Infected wound healing poses a significant global challenge due to persistent bacterial colonization, oxidative stress, and impaired angiogenesis. To address these issues, we developed a microenvironment-responsive GDQS/GA/SrHMCe hydrogel, crafted from quaternized chitosan (QCS) and dopamine-functionalized gelatin methacrylate (GelMA-DA), incorporating glycyrrhetic acid (GA) and strontium-doped hollow mesoporous...

Sodium alginate-based biomimetic crack-network Janus foam evaporator toward robust solar desalination, wastewater purification, and thermoelectric power generation.

Solar-driven interfacial evaporation is promising for sustainable freshwater production but faces challenges like salt accumulation and energy loss. Inspired by the crack networks in chameleon skin, we developed a three-dimensional Janus foam evaporator (3D-JRSMP) with engineered micro-cracks. It is constructed by anchoring recycled carbon fibers in a melamine foam scaffold integrated with a sodium alginate (SA)...

Supramolecular self-assembly of an herbal fructan and chitosan hydrogel: analysis of the fructan, gel structures and the molecular interaction mechanisms.

Polysaccharide-based supramolecular self-assembled hydrogels are promising biomaterials owing to their biocompatibility and dynamic controllability. This study was to explore the use of a polysaccharide (PSP) isolated from a well-known Chinese herb, *Polygonatum sibiricum* Red. (Huangjing) in the supramolecular self-assembled hydrogel. PSP was characterized as a neutral fructan with a backbone of $\rightarrow 1$)- β -D- Fruf -($2 \rightarrow$ and...

Antifreeze moisture-retaining coating based on ionic dextrin-hydroxyethylcellulose hydrogels: An initial insight into the structural and functional properties.

The development of antifreeze and moisture-retaining materials is important for cold-chain packaging, outdoor infrastructure, and agricultural protection. In this study, three Dextrin-hydroxyethyl cellulose (HEC) hydrogels with distinct crosslinking strategies were investigated: H1, a borate-acrylic acid covalently crosslinked network; H2, a poly(3,4-ethylenedioxythiophene)polystyrene sulfonate (PEDOT:PSS)-modified...

Hyaluronic acid-based reactive oxygen species responsive nanocomposite hydrogel for sequential drug delivery and effective prevention of postoperative abdominal adhesions.

Postoperative abdominal adhesion (PAA) affects over 90% of abdominal surgery patients and remains a huge clinical challenge. Sustained inflammation with excessive reactive oxygen species (ROS) production and subsequent peritoneal mesothelial cells (PMCs)-mesenchymal transition (MMT) are key events driving PAA formation. Herein, we engineer a ROS-responsive nanocomposite hydrogel (P/H/PBLA@PFD) for ROS-triggered...

Sandwich-structured porous foam from sodium alginate-intercalated MXene/polyethylene glycol: Photothermal conversion, phase change storage and electromagnetic interference shielding.

Integrating photothermal conversion, thermal storage, and electromagnetic interference (EMI) shielding into porous foams meets the demand for multifunctional high-performance composites. In this work, a novel composite was developed by intercalating sodium alginate (SA) between MXene and polyethylene glycol (PEG), followed by the fabrication of PEG/MXene/SA hydrogels through hydrogen bonding or coordination with...

Glucose-sensitive and 3D-printable dynamic chitosan hydrogels from boronic acid-functionalized chitosan and gallol-functionalized chitosan.

Chitosan-based hydrogels are attractive candidates for biofabrication due to their biocompatibility and bioactivity; however, their application in 3D printing is limited by poor mechanical stability, often resulting in structural collapse during printing. Here, a self-healing chitosan gallol-chitosan boronic acid (CGB) hydrogel ink is developed through the dynamic coordination of a gallol-functionalized (CG)...

An adhesive chitosan microneedle patch incorporating a Sargassum-derived polysaccharide for skin repair.

Microneedle patches have emerged as promising tools for wound repair. However, their clinical utility is limited by poor adhesion and insufficient interfacial stability, which often lead to secondary wound separation and inefficient drug delivery. Here, we report a seaweed-derived bioadhesive microneedle patch that integrates robust tissue adhesion with sustained drug release for effective skin regeneration. The...

A self-healing, pH/glucose-responsive carboxymethyl chitosan and oxidized bacterial nanocellulose hydrogel for insulin and taurine delivery in diabetic wound healing.

Chronic diabetic wounds remain challenging to treat due to persistent local insulin resistance (IR), glucose and ROS-rich microenvironment, that impair tissue repair. In this study, we developed a dual-responsive, self-healing hydrogel (CPOPI+T) designed from the polysaccharides carboxymethyl chitosan (CCS) and oxidized bacterial nanocellulose (OBC), in conjunction with phenylboronic acid (PBA) and polyvinyl alcohol...

Naturally derived polysaccharides and ganoderic acid A hydrogel with whitening potential to combat oxidative stress and melanin production caused by ultraviolet B.

The intersection of natural products and precision skin care has become a new trend in cosmetic development. This study developed a hydrogel composed of oxidized Tremella fuciformis polysaccharide, carboxymethyl chitosan, and Ganoderic acid A. The hydrogel formed a stable network through Schiff base bonds and hydrogen bonding, exhibiting excellent biocompatibility, self-healing properties, and moisturizing capacity....

Injectable double-crosslinked oxidized dextran/ gelatin-tyramine hydrogel loaded with in-situ synthesized HA on COOH-MWCNTs to enhance osteogenesis and angiogenesis.

Critical-sized bone defects remain intractable clinically due to insufficient self-healing ability. Injectable hydrogels are ideal bone repair scaffolds but suffer from weak mechanics, poor stability, and inadequate osteogenic/angiogenic functions. To solve these issues, we developed a multifunctional injectable double-crosslinked hydrogel (Odex/Gel-TA@cCNT-HA) by incorporating in-situ synthesized cCNT-HA...

A bio-nanocomposite hydrogel based on β -cyclodextrin and straw mushroom (*Volvariella volvacea*) soluble polysaccharide for the delivery of capsaicin and alleviation effect on type II diabetes mellitus.

Type II diabetes mellitus (T2DM) remains a global health challenge, necessitating innovative drug delivery strategies to improve treatment efficacy. Capsaicin (CAP), a bioactive compound with promising antidiabetic potential, is limited by its poor solubility and bioavailability. To address these limitations, a hydrogel nanocomposite system (HGNCs) based on β -cyclodextrin (β -CD) and soluble polysaccharides from...

Synergistic dual-dynamic-bonds rendered adaptable oxidized carboxymethyl cellulose/lignin/sodium alginate/poly(vinyl alcohol) biohydrogels for wound healing.

Dual dynamic bond hydrogel dressings, known for their exceptional self-healing abilities and stimulus responsiveness, offer significant benefits in promoting wound healing. Hence, a multifunctional hydrogel (OCMC/LA/AP/PVAE) was developed by forming dynamic imine and borate ester bonds among oxidized carboxymethyl cellulose, aminated lignin, modified sodium alginate, poly (vinyl alcohol) and...

Fabrication of CA-K+/P(AM-MBA)/AgNPs-K+ hydrogel with ion-electron synergistic conduction and ion-covalent dual-network enhancement and its flexible sensing performance.

Flexible strain sensors based on conductive hydrogels have attracted considerable attention for wearable electronics due to their intrinsic softness, stretchability, and biocompatibility. However, achieving a balance between mechanical robustness, high conductivity, and multifunctionality remains challenging. Herein, κ -carrageenan-based hydrogels CA-K+/P(AM-MBA)/AgNPs-K+ with a dual-network structure and...

A spatiotemporally programmed, matrix metalloproteinase-9-responsive hyaluronic acid/polyglutamate hydrogel for accelerating chronic diabetic wound healing.

In chronic diabetic wounds, immune microenvironmental dysregulation arises from hyperglycemia, accumulation of reactive oxygen species (ROS), and overexpression of matrix metalloproteinase-9 (MMP-9), leading to an imbalance in macrophage M1/M2 polarization and a dysregulated inflammatory response impeding wound healing. To address this challenge, we developed an MMP-9-activated, multimodal glycopeptide hydrogel...

3-3. 펙틴(pectin) 관련 논문 (4편)

Structural characterization and hypoglycemic activity of a novel pectic polysaccharide extracted from *Atractylodes lancea* rhizome.

Type 2 diabetes mellitus (T2DM) ranks first among global non-communicable chronic diseases, with the third highest incidence rate, making dietary intervention an effective strategy. In this study, a pectic polysaccharide named ALP-2 was isolated and characterized from *Atractylodes lancea* (Thunb.) DC., and its potential hypoglycemic activity both in vitro and in vivo was investigated. Structural studies revealed that...

Dual-configuration arabinogalactan-II synergizes with rhamnogalacturonan-I in *Lycium barbarum* pectin to construct an intestinal anti-inflammatory glycan scaffold.

Lycium barbarum pectin is a promising dietary component with potential gut health benefits, yet its precise structure-bioactivity relationship remains unclear. This study aimed to elucidate the unique architectural basis for its anti-inflammatory efficacy. Through an integrated approach combining an advanced structural analysis with in vivo and multiomics investigations, we report a novel pectin architecture. We...

A sulfated pectin polysaccharide from *Chrysanthemum morifolium* induces ferroptosis in pancreatic cancer by stabilizing TFRC.

Pancreatic ductal adenocarcinoma (PDAC) is the most malignant tumor with dismal prognosis. The chemotherapy effect of PDAC is limited and has severe side effects. Hence, identifying a natural leading compound with high efficacy against PDAC is urgently needed. *Chrysanthemum morifolium* (C. morifolium), a traditional Chinese medicinal herb with reported antitumor properties, was hypothesized to contain polysaccharides...

Characterization and encapsulation properties of pectic polysaccharides derived from *Choerospondias axillaris* fruit.

The fruit of *Choerospondias axillaris* is commonly used in the food industry for producing candies and cakes. However, limited research on polysaccharides from C. axillaris fruit has restricted their application. In this study, a homogeneous pectic polysaccharide of 22.4 kDa, designated as CAP-3, was isolated from C. axillaris fruit. Structural characterization revealed that CAP-3 exhibited a backbone composed of...

3-4. 바이오매스(biomass) 관련 논문 (4편)

Compressible lignin-enhanced holocellulose aerogel for adsorption applications.

The development of sustainable aerogels is hindered by the brittleness of inorganic varieties and the environmental burden of synthetic polymers. Herein, we have achieved the simultaneous dissolution and reconfiguration of holocellulose and lignin by employing an environmentally friendly solvent system, enabling the green fabrication of high-performance holocellulose-based aerogels incorporating all major wood...

A thermophilic GH16 β -agarase from *Salagentibacter agarivorans* enables efficient and selective production of neoagarooligosaccharides.

Agarose, a hydrophilic polysaccharide derived from red algae, is widely used in biochemical and agricultural applications, but its intrinsic gelation below $\sim 40^\circ\text{C}$ limits enzymatic conversion into high-value neoagarooligosaccharides (NAOS). Here, we report the systematic characterization of a novel thermophilic β -agarase, SaAgaD, identified from *Salagentibacter agarivorans*. SaAgaD is a 398-residue GH16 enzyme that...

Synergistic dual-dynamic-bonds rendered adaptable oxidized carboxymethyl cellulose/lignin/sodium alginate/poly(vinyl alcohol) biohydrogels for wound healing.

Dual dynamic bond hydrogel dressings, known for their exceptional self-healing abilities and stimulus responsiveness, offer significant benefits in promoting wound healing. Hence, a multifunctional hydrogel (OCMC/LA/AP/PVAE) was developed by forming dynamic imine and borate ester bonds among oxidized carboxymethyl cellulose, aminated lignin, modified sodium alginate, poly (vinyl alcohol) and...

Photothermal polypyrrole-cellulose nanofiber membranes as advanced platforms for osmotic energy conversion.

Using nanofluidic membranes to harvest osmotic energy at the interface between seawater and river water is a promising solution for mitigating energy shortages. However, the development of stimuli-responsive ion transport into nanofluidic membranes remains largely underexplored. Herein, we design a photothermally responsive nanofluidic membrane by integrating polypyrrole (PPY), TEMPO-oxidized cellulose nanofibers...

3-5. 종이/기질(paper) 관련 논문 (1편)

Construction of an ultra-sensitive surface-enhanced Raman spectroscopy substrate based on a water-stable cross-linked cyclodextrin metal-organic framework.

Surface-enhanced Raman spectroscopy (SERS) substrates based on gold nanoparticles (AuNPs) typically suffer from limited enhancement efficiency due to random AuNPs aggregation. This aggregation issue, however, can be resolved by synthesizing AuNPs using cyclodextrin metal-organic framework (CD-MOF) as a structural template-a strategy that suppresses aggregation and yields well-defined Au@CD-MOF composites with...

위 집중 분석 대상 논문들은 공통적으로 셀룰로오스나노섬유·펙틴·다당류 하이드로겔을 기반으로 하며, 화학적 개질(치환, 산화, 가교) 또는 나노복합화를 통해 기계적 강도, 자가치유성, 전도성, 습윤 반응성 등 부가 기능을 구현하는 방향으로 연구가 진행되고 있습니다. 특히 하이드로겔 계열은 상처치유·약물전달 등 생체의료 응용이 지배적이며, 셀룰로오스/바이오매스 계열은 에어로겔·필름 형태로 흡착, 에너지저장, 단열/방열 등 환경·에너지 응용으로 확장되는 경향이 뚜렷합니다.

부록: 신규 연구논문 전체 목록

총 89편 (제목 / 저자 / 게재일 / DOI)

1. Cationic architecture engineering of amphiphilic ionic liquids for synergistic cellulose processing: From molecular interaction to high-performance regenerated materials.

저자: Wang Zhao, Nuo Xu, Hao Hu, Cheng Xu 외 3인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125244 | PMID: 42002331

2. A fructan-type polysaccharide from *Lycium ruthenicum* attenuates liver fibrosis via microbiota-dependent ferroptosis inhibition.

저자: Yao Zhao, Ming Qiao, Chong Ma, Qiang Hou 외 2인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125243 | PMID: 42002330

3. Fabrication of debranched pea starch beads through gellan gum self-assembly and ionic cross-linking for enhanced digestive resistance.

저자: Lu Yang, Ziyun Zhao, Hongyan Mu, Yang Qin 외 3인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125241 | PMID: 42002328

4. Bifunctional starch nanoparticles for achieving recyclable interfacial biocatalysis within a narrow pH range.

저자: Liang Qi, Yujing Zhou, Yuling Pan, Cong Zhang 외 2인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125240 | PMID: 42002327

5. A non-equilibrium oscillatory starch derivative driven by thermally induced phase separation: Application in dynamic anti-counterfeiting and wastewater treatment.

저자: Wenjun Yan, Tingmin Ran, Kaixiang Wang, Ayyaz Ahmad 외 2인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125239 | PMID: 42002326

6. Preparation and characterization of starch nanoparticles: Effects of various natural deep eutectic solvents.

저자: Raoyi Xiong, Liping Deng, Xiuting Hu, Shunjing Luo 외 2인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125238 | PMID: 42002325

7. Starch fine structure predicts glycemic index variation in whole-grain rice.

저자: Putlih Adzra Pautong, Rhowell Jr Navarro Tiozon, Reuben James Q Buenafe, Ana Rose Ramos-Castrosanto 외 6인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125236 | PMID: 42002323

8. Decreased gelatinization and retrogradation promoting cold storage stability in glutinous rice via low SSIIa activity-mediated structural modifications of amylopectin.

저자: Cheng Liang, Dan Liu, Yuesi Bu, Chiyu Wang 외 7인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125235 | PMID: 42002322

9. Unraveling the binding mechanism and enhanced performance of indigo/ β -cyclodextrin inclusion complex: An integrated experimental and computational study.

저자: Shu Tang, Zixuan Tao, Cen Li, Chunling Zheng | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125234 | PMID: 42002321

10. Uniaxial tensile strain-insensitive and zero-birefringence cellulose acetate films via atom-economic isocyanate modification.

저자: Zhouhang Lei, Wenhao Guo, Xingyue Fang, Zhengqi Tan 외 3인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125232 | PMID: 42002320

11. Lightweight and eco-friendly methyl cellulose foams featuring moisture-induced form interlocking: A versatile alternative to conventional plastic foams.

저자: Rufe Ge, Longtai Wang, Yanzhen Lou, Yan Zhou 외 5인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125231 | PMID: 42002319

12. Laponite-reinforced marine bioinspired injectable self-healing adhesive chitosan-based hydrogel for infected wounds.

저자: Ruirui Guan, Yuhui Su, Shanshan Li, Shuo Yang 외 2인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125230 | PMID: 42002318

13. Impact of polyampholyte macromolecular structure on the conformation of chitosan derivatives.

저자: Jordyn Ann Howard, Juan Felipe Salazar Ariza, Agnès Crépet, Alexander Zaderko 외 5인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125229 | PMID: 42002317

14. A sulfated manno-glucuronan ameliorates β -cell dysfunction in type 2 diabetes by targeting ALDH1A3.

저자: Wenjing Zhang, Fuming Zhang, Xiaoting Zou, Nan Wu 외 6인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125227 | PMID: 42002316

15. Skin-inspired cellulose-based hydrogel design with hydrophobic-conductive synergy for enhanced electrochemical nanoplastic removal.

저자: Wenqin Zhong, Yehan Tao, Jialu Song, Jian Du 외 5인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125298 | PMID: 42002313

16. Sodium alginate modified by dopamine multifunctional injectable hydrogel with photothermal, antibacterial, and hemostatic properties for accelerating infected wound repair.

저자: Xinbo Ma, Xiaonan Huang, Xuelian Zhang, Jiani Zhang 외 6인 | 게재일: 2026-Jun-15 | DOI: 10.1016/j.carbpol.2026.125188 | PMID: 42002312

17. On-demand amorphous calcium phosphate-mediated regeneration of dentin and enamel using ion-regulated hydroxypropyl methylcellulose and carboxymethyl chitosan hydrogels.

저자: Zihan Cui, Zhiwen Liu, Zhixin Zhang, Danni Wang 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125170 | PMID: 41943389

18. Chitosan-gelatin hybrid hydrogel incorporating strontium-doped ceria nanoparticles for enhanced wound healing and mitochondrial function recovery.

저자: Peng Zhang, Ruixun Chen, Xinmiao Luo, Zuqiang Shi 외 7인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125165 | PMID: 41943387

19. Boosting longan seed starch-flavonol complex formation via cold plasma: Structural, functional, and mechanistic insights.

저자: Rui Huang, Rui He, Xiaoyu Lin, Jiao Chen 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125163 | PMID: 41943385

20. Sodium alginate-based biomimetic crack-network Janus foam evaporator toward robust solar desalination, wastewater purification, and thermoelectric power generation.

저자: Xin Wang, Xiaowen Cui, Xiao Sun, Yicheng Wang 외 10인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125162 | PMID: 41943384

21. Two low molecular weight-polysaccharides isolated from the edible fungus *Sarcodon imbricatus*: Isolation, characterization, immunomodulatory and intestinal barrier-protective effects.

저자: Tak Ho Lo, Xiyin Zheng, Kunzhui Chen, Jiaxian He 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125161 | PMID: 41943383

22. Supramolecular self-assembly of an herbal fructan and chitosan hydrogel: analysis of the fructan, gel structures and the molecular interaction mechanisms.

저자: Yong Zhu, Fengyan Chen, Bingmin Wu, Yishan Tong 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125160 | PMID: 41943382

23. Hyaluronic acid-based supramolecular peptide-drug conjugate prodrug nanosystems for precise multiple myeloma therapy.

저자: Xiangyu Deng, Qili Huang, Li Cao, Ran Luo 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125159 | PMID: 41943381

24. Bio-based gellan gum electrolytes reinforced with SiO₂ Nanofillers: Enhanced ion transport for solid-state zinc-ion batteries.

저자: T Mohana Selvi, V Mareeswaran, S Brindha, P Arjunan 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125158 | PMID: 41943380

25. Optimization of the quality and physicochemical properties of selenium-enriched wheat fresh noodles by incorporating a wheat starch-based complex improver.

저자: Zhi-Hong Zhang, Huiyue Wang, Kaiwei Zhang, Xiaolan Li 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125156 | PMID: 41943378

26. Processing driven multi-scale structures dominate the anti-digestibility and peptide release behavior of starch-proteolytic peptide complexes.

저자: Xiang Huang, Yuxuan Hu, Yue Li, Song Zhu 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125154 | PMID: 41943377

27. Antifreeze moisture-retaining coating based on ionic dextrin-hydroxyethylcellulose hydrogels: An initial insight into the structural and functional properties.

저자: Nurfarhanim Abu Bakar, Khairul Ikhwan Mali, Huda Salah Kareem, S R Majid 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125155 | PMID: 41943376

28. Hierarchical assembly and digestive resistance of starch-unsaturated fatty acid complexes regulated by ultrasound-assisted high-pressure microfluidization.

저자: Bo Zheng, Hui Wen, Xingwei Xiang, Bin Zheng | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125153 | PMID: 41943375

29. 222-nm far-UVC/H₂O₂ photodegradation as a versatile and effective tool for depolymerizing starch-rich materials and its application as a sustainable approach to starch liquefaction.

저자: Zhenhui Jin, Lei Xing, Ping Dong, Yanli Shang | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125152 | PMID: 41943374

30. In vitro mammalian intestine mucosal enzymatic digestion of starch spherulites formed from debranched waxy maize and high-amylose maize starches.

저자: Myeongsu Jo, Yong-Cheng Shi | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125151 | PMID: 41943373

31. Oligochitosan and human serum albumin: Focus on compatibility, protein stability, and antibacterial activity in isotonic solution.

저자: Evgeniya A Bezrodnykh, Yuriy A Antonov, Irina L Zhuravleva, Sergey N Kulikov 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125150 | PMID: 41943372

32. Geometry-matched chelation enhances Li⁺ diffusion on electrodes binded by modified carboxymethyl cellulose.

저자: Zi Ye, Xi Yang, Wenhan Chen, Xinye Li 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125149 | PMID: 41943371

33. Hyaluronic acid-based reactive oxygen species responsive nanocomposite hydrogel for sequential drug delivery and effective prevention of postoperative abdominal adhesions.

저자: Yujun Gong, Danni Xiao, Tao Zhang, Xianmin Shi 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125148 | PMID: 41943370

34. High-molecular weight hyaluronic acid protects against colitis by remodeling microbiota and restoring barrier function.

저자: Xinwei Gao, Hao Huang, Litao Hu, WeiQuan Ding 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125145 | PMID: 41943367

35. Construction of an ultra-sensitive surface-enhanced Raman spectroscopy substrate based on a water-stable cross-linked cyclodextrin metal-organic framework.

저자: Meimei Guo, Tahirou Sogore, Jin Huang, Xinyu Liao 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125144 | PMID: 41943366

36. Sandwich-structured porous foam from sodium alginate-intercalated MXene/polyethylene glycol: Photothermal conversion, phase change storage and electromagnetic interference shielding.

저자: Chengzhe Niu, Mingyu Li, Gaofei Pan, Jingwen Lu 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125143 | PMID: 41943365

37. λ -Carrageenan inhibits African swine fever virus infection potentially through its interaction with the viral envelope protein CD2v.

저자: Kun Han, Limeng Sun, Can Jin, Siqi Wan 외 20인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125142 | PMID: 41943364

38. Glucose-sensitive and 3D-printable dynamic chitosan hydrogels from boronic acid-functionalized chitosan and gallol-functionalized chitosan.

저자: Ting-Yu Chang, Tsai-Yu Chen, Qian-Pu Cheng, Shan-Hui Hsu | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125140 | PMID: 41943362

39. Sustainable chitosan composite foams: Synergistic hydroxyapatite nanowires / phytic acid-melamine hybridization for reliable fire warning and efficient flame suppression.

저자: Zhirong Xu, Pengyu Xing, Yong Pan, Yuling Xiao | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125137 | PMID: 41943360

40. Transitions in structural, functional, and digestibility properties of durum wheat starches with normal, intermediate, and high amylose levels.

저자: Yikai Ren, Fan Cheng, Yan Cai, Michael Reimer 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125136 | PMID: 41943359

41. An adhesive chitosan microneedle patch incorporating a Sargassum-derived polysaccharide for skin repair.

저자: Qianqian Ouyang, Hongjiao Qu, Weiyan Quan, Yongkai Tang 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125135 | PMID: 41943358

42. A self-healing, pH/glucose-responsive carboxymethyl chitosan and oxidized bacterial nanocellulose hydrogel for insulin and taurine delivery in diabetic wound healing.

저자: Md Fardin Khan, Luyong Zhang, Dongyan Qiang, Zhenhai Zhang 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125133 | PMID: 41943357

43. Hydroxypropyl cellulose/thermoplastic polyurethane blend films with reversible optical performance triggered by water.

저자: Yimin Yao, Yang Zhou, Glenn A Spiering, Isabela Trindade Coutinho 외 7인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125128 | PMID: 41943356

44. Folic acid functionalized exopolysaccharide integrated into a coordination polymer for targeted and controlled drug release.

저자: Jose Paul, Rajapriya Govindaraju, Jongsung Kim | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125127 | PMID: 41943355

45. Naturally derived polysaccharides and ganoderic acid A hydrogel with whitening potential to combat oxidative stress and melanin production caused by ultraviolet B.

저자: Ting-Ting Ye, Tonghe Liu, Mingzhen Liu, Xudong Xu 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125126 | PMID: 41943354

46. Pilot-scale production, purification, and structural characterization of a glycosaminogalactan exopolysaccharide from *Lactiplantibacillus pentosus*.

저자: Ahmad Tsjokajev, Gordon Jacob Boehlich, Victor Daisuke Kietzmann, Lars Fredrik Moen 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125125 | PMID: 41943353

47. Compressible lignin-enhanced holocellulose aerogel for adsorption applications.

저자: Jiahao Yang, Wenchao Jia, Lu Wu, Lingzhi Huang 외 6인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125116 | PMID: 41943352

48. A mussel-inspired universal strategy for fabricating dual-bactericidal dressings to eradicate biofilms and accelerate infected wound healing.

저자: Guang Chen, Ruijie Xu, Shuang Li, Xiaoxue Chen 외 6인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125113 | PMID: 41943350

49. Structural characterization and hypoglycemic activity of a novel pectic polysaccharide extracted from *Atractylodes lancea* rhizome.

저자: Xiaozhen Wang, Weiyi Zhao, Xia Li, Lanping Guo 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125096 | PMID: 41943349

50. Injectable double-crosslinked oxidized dextran/ gelatin-tyramine hydrogel loaded with in-situ synthesized HA on COOH-MWCNTs to enhance osteogenesis and angiogenesis.

저자: Xiaoyuan Liu, Ling Liu, Long Yang, Haotian Bai 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125052 | PMID: 41943346

51. Dual-configuration arabinogalactan-II synergizes with rhamnogalacturonan-I in *Lycium barbarum* pectin to construct an intestinal anti-inflammatory glycan scaffold.

저자: Guoqiang Li, Yajun Qiao, Qiannan Wang, Xingfang Zhang 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.124914 | PMID: 41943336

52. A bio-nanocomposite hydrogel based on β -cyclodextrin and straw mushroom (*Volvariella volvacea*) soluble polysaccharide for the delivery of capsaicin and alleviation effect on type II diabetes mellitus.

저자: Wei Hu, Zhihong Zeng, Lanming Gou, Xiaoling Luo 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2025.124857 | PMID: 41943335

53. Acid-treated sodium alginate-based composite gel microspheres toward methylene blue removal: Fabrication and mechanistic insights.

저자: Hongxing Shi, Qian Zhao, Mingjie Han, Yin Li 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125233 | PMID: 41943332

54. Chemical structure and P-selectin inhibition of two linear and highly regular homofucans from the brown alga *Fucus vesiculosus*.

저자: Linxia Chen, Dilihumaer Ruzemaimaiti, Lige Cui, Guangyu Zhu 외 10인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125228 | PMID: 41943331

55. A thermophilic GH16 β -agarase from *Salegentibacter agarivorans* enables efficient and selective production of neoagarooligosaccharides.

저자: Wenhui Nian, Miao Zhao, Alejandra Moenne, Tang Li 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125226 | PMID: 41943330

56. Anti-aging effects of low-molecular-weight polysaccharide PSP-1-1 from steamed *Polygonatum sibiricum* via gut microbiota modulation.

저자: Chen Zhang, Hongru Lin, Congmin Wei, Xue Long 외 6인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125225 | PMID: 41943329

57. A sulfated pectin polysaccharide from *Chrysanthemum morifolium* induces ferroptosis in pancreatic cancer by stabilizing TFRC.

저자: Zixuan Wang, Youning Jiang, Yilin Wang, Shengjie Wu 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125224 | PMID: 41943328

58. Ethanol-assisted alkyl ketene dimer surface modification for enhanced moisture stability of starch-based foams.

저자: Yuzhuo Huang, Shaobo Zhang, Long Yu, Qingfei Duan 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125222 | PMID: 41943327

59. Co-solvency phenomenon and thermodynamic mechanism of cyanoethyl cellulose in acetone-water binary solvent.

저자: Zhenyu Qiao, Hao Wu, Jinping Zhou | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125221 | PMID: 41943326

60. Synergistic dual-dynamic-bonds rendered adaptable oxidized carboxymethyl cellulose/lignin/sodium alginate/poly(vinyl alcohol) biohydrogels for wound healing.

저자: Zhu Shan, Bo Jiang, Sehrish Manan, Peng Wang 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125220 | PMID: 41943325

61. Low-molecular-weight ferulic oligosaccharides derived from rice bran enhance the inhibitory effect on rice starch digestion by attenuating self-aggregation.

저자: Mengwei Chang, Guangsu Zhu, Kunlun Liu | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125219 | PMID: 41943324

62. Fabrication of CA-K+/P(AM-MBA)/AgNPs-K+ hydrogel with ion-electron synergistic conduction and ion-covalent dual-network enhancement and its flexible sensing performance.

저자: Yihan Yan, Shenghua Lv, Yuanyuan Qiang, Jialong She 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125218 | PMID: 41943323

63. A spatiotemporally programmed, matrix metalloproteinase-9-responsive hyaluronic acid/polyglutamate hydrogel for accelerating chronic diabetic wound healing.

저자: Chunxiao Wang, Qing Liu, Ning Wang, YINUO Fan 외 6인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125211 | PMID: 41943322

64. Physicochemical properties and molecular weight-dependent gut microbiota modulation of inulin-type fructans from *Codonopsis bulleyana*.

저자: Debing Pu, Tao Liang, Xuan Li, Wanqi Xing 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125217 | PMID: 41943321

65. Robust, hydrophobic and antimicrobial TiO₂-decorated bamboo cellulose nanofiber aerogel with honeycomb porous structure for highly efficient radiative cooling.

저자: Xinmeng Luo, Yingying Lu, Jinlong Yu, Zhiyong Qin 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125215 | PMID: 41943320

66. Levan-derived oligosaccharides (LOS) prime barley against fungal pathogens through asparagine metabolism.

저자: Dan Lian, Yanjing Li, Jiabin Ma, Junjie Zhou 외 9인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125213 | PMID: 41943319

67. Sustainable and reusable antibacterial aerogels constructed from surface-quaternized cellulose nanofibrils for effective oil-water separation.

저자: Pengfei Wang, Linghua Xu, Lianjie Zhao, Xuefei Cao 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125212 | PMID: 41943318

68. Synergistic effects of heat and shear during high-pressure homogenization on corn starch with different amylose contents: Multi-scale structural evolution to pasting properties.

저자: Fuhua Xie, Huabing Zhai, Yongmei Mo, Yaqi Zhong 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125210 | PMID: 41943317

69. Production of electrospun cellulose fibres via water-assisted rapid deacetylation: Experimental and molecular modelling insights.

저자: Shameek Vats, Ali Khodayari, Clément Mugemana, Stefan Spirk 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125204 | PMID: 41943311

70. Improving oleoyl hyaluronan self-assembly and functional properties using zinc ions.

저자: Jaroslav Sita, Martina Štěpánková, Kateřina Lehká, Vojtěch Pavlík 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125203 | PMID: 41943310

71. Antioxidant functionalization of pullulan with ferulic acid using enzymatic catalysis.

저자: Koceila Boundaoui, Xavier Falourd, Corinne Loutelier-Bourhis, Tony Varacavoudin 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125202 | PMID: 41943309

72. Directed synthesis, characterization and rheological properties of an ultrahigh molecular weight acetylated-gellan gum.

저자: Yue Ming, Zhuangzhuang Shi, Weilong Wang, Jue Li 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125201 | PMID: 41943308

73. Ultralight core-shell polypyrrole@bacterial cellulose aerogel for broadband electromagnetic wave absorption and efficient heat insulation.

저자: Long Zhao, Haoxiang Zhai, Yongxin Qian, Xuefei Xu 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125200 | PMID: 41943307

74. A novel xylosylated fucoglucuronan in *Penium* reveals structural parallels to rhamnogalacturonan-I and its broad evolutionary footprint in lower plants.

저자: Li Tan, Varughese Mulamootil, Parastoo Azadi | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125199 | PMID: 41943306

75. Properties of agar aerogels: Effect of concentration and ageing time.

저자: Nika Atel ek Hozjan, Gabrijela Horvat, Lara Gibowsky, Pavel Gurikov 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125198 | PMID: 41943305

76. Preparation of polypyrrole-ulvan nanoparticles and bioactive properties of multifunctional polypyrrole-ulvan/chitosan dressings for enhanced wound healing.

저자: Trong-Ming Don, Hung-Chih Tai, Ying-Chun Chen, Yi-Cheng Huang | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125196 | PMID: 41943303

77. Waxy gene deficiency shapes the multi-level wheat starch structure and noodle quality.

저자: Fada Han, Shuo Zhang, Liqiang Song, Liang Chen 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125195 | PMID: 41943302

78. Enzymatic conversion of α -1,4 to α -1,3 linkages by glyco transferase for tailoring cassava starch structure and digestibility.

저자: Tianyu Ji, Jiayu Cui, Yiming Wei, Shiyao Yu 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125194 | PMID: 41943301

79. Antibacterial chitosan bioplastics with enhanced water stability via electro-assembly followed by direct thermal processing.

저자: Ruizhe Rao, Mengyi Liu, Xiao Yu, Xiaoyang Li 외 2인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125190 | PMID: 41943300

80. Multi-scale characterization and artificial neural network-based modeling identify leachate composition as a primary factor of cooked rice palatability.

저자: Mingyo Ha, Dong-Hwa Cho, Hyun-Jung Chung | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125187 | PMID: 41943298

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저자: Zhijie Zhu, Zijun Yu, Wenjing Li, Juan Tang 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125186 | PMID: 41943297

82. Synergistic induction of immunogenic cell death and tumor microenvironment remodeling by biotinylated marine polysaccharide nanoparticles for enhanced colorectal cancer therapy.

저자: Hui Wang, Haoyu Yu, Yan Yang, Mengying Fu 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125185 | PMID: 41943296

83. Integrated in silico design of self-assembled herbal bioactives-polysaccharides nanoformulation for targeted therapy against gastric pathogen *Helicobacter pylori*.

저자: Chuqiu Zhang, Haobo Chen, Gengyuan Yu, Yuqian Lai 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125184 | PMID: 41943295

84. Virus-inspired *Ganoderma lucidum* polysaccharide-functionalized adjuvant with pseudo-zwitterionic interface for potentiated nasal immunization.

저자: Xu-Han Liu, Zhe Zhai, Ling-Ling Tao, Nan-Yu Chen 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125183 | PMID: 41943294

85. Characterization and encapsulation properties of pectic polysaccharides derived from *Choerospondias axillaris* fruit.

저자: Hao Liu, Ruolan Du, Hongxiang Li, Wenbo Zhao 외 4인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125175 | PMID: 41943292

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저자: Wen Yin, Zhuojun Yan, Ziyuan Wang, Zhenlei Zhou | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125174 | PMID: 41943291

87. Pros and cons of wood cellulose binders for 4.6 V LiCoO₂ lithium-ion batteries.

저자: Yunyu Tong, Xinyue Zheng, En Chen, Zihan Guo 외 3인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125173 | PMID: 41943290

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저자: Wenlong Zhang, Kai Zhao, Mehraj Ahmad, Zhouyue Li 외 1인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125172 | PMID: 41943289

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저자: Yu Meng, Qin Lu, Yangbing Wen, Zhiping Chen 외 5인 | 게재일: 2026-Jun-01 | DOI: 10.1016/j.carbpol.2026.125171 | PMID: 41943288